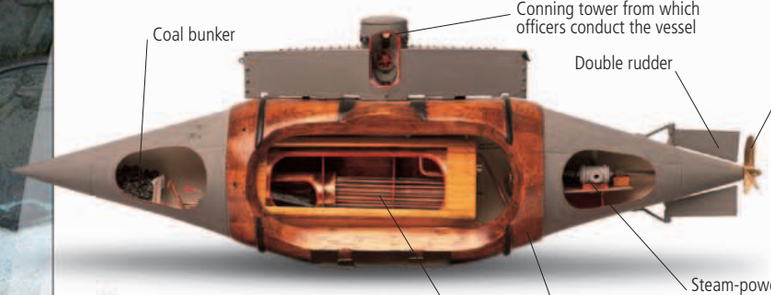
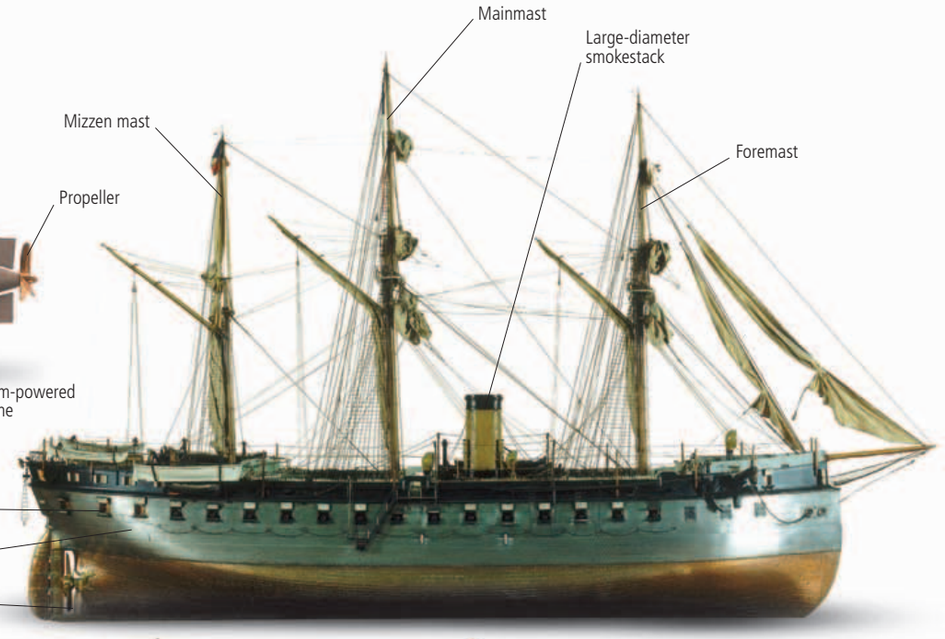
IRON AND STEEL WARSHIPS

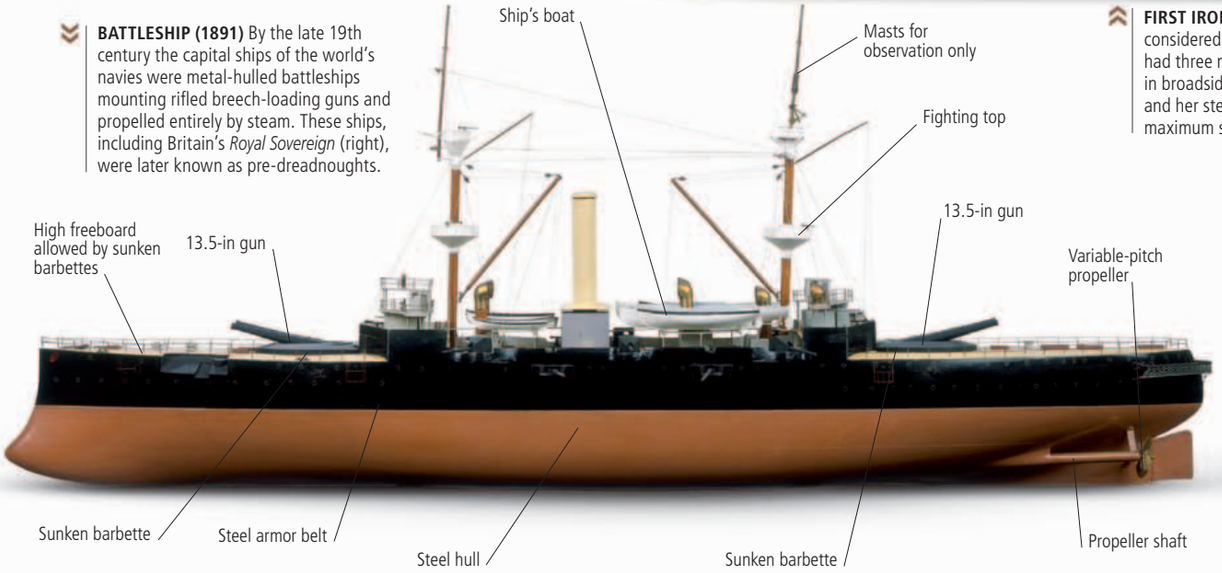


EARLY SUBMARINE (1879) An experimental British submersible, *Resurgam* (above) was steam powered. When submerged the vessel was powered by steam stored in pressurized tanks. She sank in 1880.



FIRST IRONCLAD (1859) The French vessel *Gloire* is considered the first true ironclad warship. She still had three masts, a wooden hull, and guns arranged in broadside, but her hull was clad with iron armor and her steam-driven screw propeller gave her a maximum speed of 13 knots.

BATTLESHIP (1891) By the late 19th century the capital ships of the world's navies were metal-hulled battleships mounting rifled breech-loading guns and propelled entirely by steam. These ships, including Britain's *Royal Sovereign* (right), were later known as pre-dreadnoughts.





THE DESIGN OF WARSHIPS in the 19th century was revolutionized by the adoption of steam power, the replacement of wood by iron and steel in ship construction, and the development of rifled turret guns firing explosive shells. Change was at first quite slow. The first steam-powered fighting ship was the *Demologos*, an American paddle-wheel-driven harbor defense vessel of 1814, and the first iron-hulled ships emerged in the 1820s. But until the mid-19th century major navies still consisted primarily of wooden sailing ships. The advent of the screw propeller as an alternative to the paddle wheel was a crucial step forward (the first screw-driven warship was launched in 1843) and by the 1850s almost all new-built warships had steam engines as well as sails. The universal adoption of metal construction took longer. In the 1850s ships with hulls of wood or iron were covered with thick iron plates, creating the "ironclad". In the 1870s steel began to be used both for hulls and armor, while sails finally disappeared as marine steam engines became more effective. This revolution in ship construction and propulsion was matched by changes in armament. Explosive shells replaced solid shot. Firing in broadside was gradually abandoned in favor of a much smaller number of large breech-loading guns mounted in revolving turrets—the effectiveness of a turret gun having been demonstrated by the American *Monitor* of 1861. The deployment of torpedoes created new problems for naval tacticians, and gave designers a chance to create new kinds of small, fast torpedo attack craft.



Explosive shell with sabot
This explosive shell from the American Civil War is attached to a wooden sabot. The shell was placed sabot-first into the muzzle to keep the shell's fuze centered; the fuze was ignited by the powder charge.

UNEASY TRANSITION

Some of the large warships built during the period of transition from wood and sail to steam and steel were ungainly and even unseaworthy. The positioning of guns, funnels, and masts (still needed for observation posts) often involved unsatisfactory compromises, as did the trade-off between armor and performance.